SupportShift: Matérn Range Distortion under Local Averaging with Directional and Finite-Library Guarantees

Sasha Cui
Independent researcher

Draft of August 2, 2026

Abstract

Spatial measurements often represent local averages, but are subsequently fitted as if they were point observations. This changes the population target of covariance estimation, not merely its finite-sample bias. We study a stationary Matérn Gaussian field in $\mathbb{R}^d$, observed through a known compact symmetric averaging kernel of bandwidth h. For a fixed nonzero-lag Gaussian two-point likelihood with known smoothness ν, we characterize the exact pseudo-variance and pseudo-decay and derive their small-support expansions. The naive inverse-range shift is of order $h^{2\nu}$ for $0 < \nu < 1$, $h^2 \log(1/h)$ at $\nu = 1$, and h^2 for $\nu > 1$. Explicit coefficients show that naive point-support fitting underestimates inverse range—and therefore overestimates spatial range—for every fixed $\nu > 0$ and all sufficiently small h. For anisotropic support, an all-smoothness directional contrast isolates the order-h^2 difference between lag directions. We also give an explicit finite-library Gaussian likelihood certificate based on standard quadratic-form concentration. It allows arbitrary dependence among the p coordinates of each spatial field; under uniform relative spectral control its radius is $O\{\sqrt{\log(M)/(Np)} + \log(M)/(Np)\}$ for N independent fields and M covariance candidates. Deterministic quadrature verifies the phase law and directional coefficient. Replicated-field simulations verify the likelihood certificate and show that increasing information can concentrate a naive estimator around a wrong KL target. The covariance transformation and concentration inequality are established tools; the narrower contribution is their theorem-linked use with the explicit Matérn support-distortion law.

1 Introduction

Spatial values recorded on pixels, grid cells, administrative regions, or after kernel preprocessing are local aggregates of an underlying field. The change-of-support literature models such observations by integrating the latent process over their supports [3, 7]. In practice, however, smoothed values are often assigned to declared centers and a point-support covariance is fitted. Relative to the latent parameters, this is a support-misinterpreted model. A single two-point covariance can still be matched exactly by different variance and decay parameters; a multi-lag likelihood is genuinely misspecified and selects a Kullback–Leibler (KL) projection [2].

That qualitative conclusion is familiar, but it does not answer three basic questions. How fast does the target move as the support shrinks? Does the rate depend on Matérn smoothness? Is the direction of the range error stable across dimension and averaging kernels? These questions matter because the Matérn covariance changes regularity at the origin as ν crosses one. A symmetric kernel cancels first-order Taylor terms at nonzero lags, but it cannot cancel a cusp or fractional power at the origin.

Temporal aggregation of autoregressive and Ornstein–Uhlenbeck models is classical [1, 15], and integrated Ornstein–Uhlenbeck inference has been studied directly [5]. Consequently, a one-dimensional exponential calculation alone is not a novelty claim. Reviews and geostatistical methods for incompatible supports establish the block-to-block covariance and area-to-point prediction problems [8, 10]. Most closely, Fuentes [6] constructs a likelihood for block averages in terms of an underlying point covariance and studies a Matérn example. That work correctly models the support; it does not derive the pseudo-range selected when block averages are instead treated as points. Generic misspecified-GP KL theory [2] supplies the projection framework but not the support-specific coefficient or sign. Gaussian quadratic-form concentration and finite-class union bounds are also standard [9, 11]; we use them to make the simulation certificate match the fitted covariance library exactly. Our principal mathematical contribution is the explicit all-smoothness, d-dimensional fixed-lag pseudo-parameter expansion. In particular:

1. We derive a phase law for the naive Matérn decay target: orders $h^{2\nu}$, $h^2 \log(1/h)$, and h^2 in the rough, threshold, and smooth regimes.

2. We give explicit coefficients. Origin expansions, together with a Bessel-function identity in the smooth regime, prove that normalized correlation increases, hence inverse range decreases, for every $\nu > 0$ and any compact symmetric kernel when h is sufficiently small.

3. We isolate the order-h^2 contrast between two lag directions under anisotropic support. The contrast holds for every $\nu > 0$, even when it is lower order than the direction-independent rough-field shift.

4. We distinguish the continuous interior theorem from discrete, row-normalized, and boundary-truncated smoothing. We then compute finite-grid full-likelihood KL targets and compare naive and support-aware fits using only synthetic data.

5. For a deterministic finite covariance library, we state a simultaneous normalized-likelihood certificate and its empirical-risk consequence. This supporting result uses independent replicated fields, not independent spatial coordinates.

The covariance transformation $SKS^\top$, block kriging, generic KL projection under misspecification, temporal aggregation, and the concentration machinery are prior work; they are not claimed as contributions. Recent finite-grid Matérn likelihood work also treats discretization and edge effects without studying the target induced by a preprocessing smoother [13]. Our narrower focus is the smoothness-dependent target displacement caused by ignoring observation support, its directional coefficient, and a benchmark in which the probability guarantee and fitted library are the same mathematical objects.

2 Model and pseudo-target

2.1 Latent field and local support

Let $Y = \{Y(t) : t \in \mathbb{R}^d\}$ be a mean-zero stationary Gaussian field with

$$\operatorname{Cov}\{Y(t), Y(t+r)\} = C(r) = v\mathcal{M}_\nu(\alpha\|r\|), \qquad v > 0, \quad \alpha > 0, \quad \nu > 0, \tag{1}$$

where

$$\mathcal{M}_\nu(x) = \frac{2^{1-\nu}}{\Gamma(\nu)} x^\nu K_\nu(x), \qquad \mathcal{M}_\nu(0) = 1. \tag{2}$$

Thus α is an inverse range; larger α means shorter dependence. This convention must not be mixed with the common length-scale parameterization $\alpha = \sqrt{2\nu}/\ell$.

Let k be a symmetric probability density supported on $B(0, L) \subset \mathbb{R}^d$, let $k_h(u) = h^{-d} k(u/h)$, and define the mean-square convolution

$$Z_h(t) = \int_{\mathbb{R}^d} k_h(u) Y(t-u)\, du. \tag{3}$$

If $U, V \overset{\text{iid}}{\sim} k$ and $D = U - V$, stationarity and Fubini's theorem give the exact smoothed covariance

$$C_h(r) = v\mathbb{E}[\mathcal{M}_\nu\{\alpha\|r + hD\|\}]\,. \tag{4}$$

Write

$$\Sigma_k = \mathbb{E}(UU^\top), \quad T_k = \operatorname{tr}(\Sigma_k), \quad m_q = \mathbb{E}\|D\|^q, \tag{5}$$

so that $m_2 = 2T_k$. We assume $T_k > 0$.

2.2 Naive Gaussian pair fit

Fix $r = Re$, with $R > 0$ and $\|e\| = 1$. The smoothed pair has variance $C_h(0)$ and correlation

$$\rho_h(r) = \frac{C_h(r)}{C_h(0)}. \tag{6}$$

A naive point-support model with the correct known ν but unknown variance and decay uses covariance

$$v^\dagger \begin{pmatrix} 1 & \mathcal{M}_\nu(\alpha^\dagger R) \\ \mathcal{M}_\nu(\alpha^\dagger R) & 1 \end{pmatrix}. \tag{7}$$

The true two-by-two covariance belongs to this family. Therefore its Gaussian KL minimizer is exactly

$$v_h^\dagger = C_h(0), \qquad \mathcal{M}_\nu(\alpha_h^\dagger R) = \rho_h(r). \tag{8}$$

Existence follows from $0 < \rho_h(r) < 1$ for $r \neq 0$. Positivity follows from $k \geq 0$ and $\mathcal{M}_\nu > 0$. Strict inequality follows because the smoothed field has spectral density $|\widehat{k}(h\omega)|^2 f_\nu(\omega)$, which is positive on a neighborhood of the origin, while $e^{i\omega^\top r}$ is not identically one there. Uniqueness follows from

$$\mathcal{M}_\nu'(x) = -\frac{2^{1-\nu}}{\Gamma(\nu)} x^\nu K_{\nu-1}(x) < 0, \qquad x > 0. \tag{9}$$

For completeness, consider a fixed lattice, a lag r belonging to that lattice, and increasing rectangular Følner domains. If the parameter space is compact, the target is unique and interior, correlations are bounded away from one, and the pair log likelihood is continuous with an integrable quadratic envelope, the ergodic theorem gives the uniform law needed for pair-composite consistency at $(v_h^\dagger, \alpha_h^\dagger)$. Overlapping pairs are dependent, so uncertainty requires a Godambe or long-run covariance, not an independence Hessian. We do not use fixed-domain asymptotics to claim separate consistency of variance and range; such a claim would conflict with Matérn microergodicity results [14, 16].

3 Small-support phase law

For $z = \alpha R$, define the directional kernel variance $a_k(e) = e^\top \Sigma_k e$ and

$$B_{\nu,k}(r) = \alpha^2 \left[a_k(e)\mathcal{M}_\nu''(z) + \{T_k - a_k(e)\} \frac{\mathcal{M}_\nu'(z)}{z} \right]. \tag{10}$$

Proposition 1 (Fixed nonzero lag)**.** *For r in a compact annulus bounded away from zero and $h \le \|r\|/(4L)$,*

$$\frac{C_h(r)}{v} = \mathcal{M}_\nu(\alpha\|r\|) + h^2 B_{\nu,k}(r) + O(h^4). \tag{11}$$

The remainder is uniform when α also ranges over a compact subset of $(0,\infty)$.

The variance behaves differently because the Matérn function is expanded at the origin. Put

$$c_\nu = -\frac{\Gamma(-\nu)}{2^{2\nu}\Gamma(\nu)} = \frac{\Gamma(1-\nu)}{\nu 2^{2\nu}\Gamma(\nu)} > 0, \qquad 0 < \nu < 1. \tag{12}$$

Theorem 1 (Matérn support phase law)**.** *Fix $\nu > 0$. Under the preceding assumptions, as $h \downarrow 0$,*

$$\frac{C_h(0)}{v} = 1 - c_\nu m_{2\nu} (\alpha h)^{2\nu} + O(h^2), \qquad 0 < \nu < 1, \tag{13}$$

$$\frac{C_h(0)}{v} = 1 - \frac{m_2}{2} (\alpha h)^2 \log\{1/(\alpha h)\} + O(h^2), \qquad \nu = 1, \tag{14}$$

$$\frac{C_h(0)}{v} = 1 - \frac{m_2}{4(\nu-1)} (\alpha h)^2 + o(h^2), \qquad \nu > 1. \tag{15}$$

Moreover, the naive decay target in (8) satisfies

$$\alpha_h^\dagger - \alpha = \frac{\mathcal{M}_\nu(z) c_\nu m_{2\nu} \alpha^{2\nu}}{R\mathcal{M}_\nu'(z)} h^{2\nu} + O\left(h^{\min(4\nu,2)}\right), \qquad 0 < \nu < 1, \tag{16}$$

$$\alpha_h^\dagger - \alpha \sim \frac{m_2 \mathcal{M}_1(z)\alpha^2}{2R\mathcal{M}_1'(z)} h^2 \log(1/h), \qquad \nu = 1, \tag{17}$$

$$\alpha_h^\dagger - \alpha = \frac{A_{\nu,k}(r)}{R\mathcal{M}_\nu'(z)} h^2 + \mathcal{R}_\nu(h), \qquad \nu > 1, \tag{18}$$

where

$$A_{\nu,k}(r) = \alpha^2 \{(2\nu - 2)a_k(e) + T_k\} G_\nu(z) > 0, \tag{19}$$

$$G_\nu(z) = \frac{\mathcal{M}_\nu(z)}{2\nu - 2} + \frac{\mathcal{M}_\nu'(z)}{z} = \frac{2^{1-\nu}}{\Gamma(\nu)} \frac{z^\nu K_{\nu-2}(z)}{2\nu - 2} > 0, \tag{20}$$

and $\mathcal{R}_\nu(h) = O(h^{2\nu})$ for $1 < \nu < 2$, $O\{h^4 \log(1/h)\}$ for $\nu = 2$, and $O(h^4)$ for $\nu > 2$. For fixed ν, these remainders are uniform for α in a compact subset of $(0,\infty)$ and r in a compact annulus bounded away from zero. They are not uniform as ν approaches one or two.

Every leading coefficient in (16)–(18) is negative because $\mathcal{M}_\nu'(z) < 0$. Thus, for sufficiently small positive h,

$$v_h^\dagger < v, \qquad \rho_h(r) > \mathcal{M}_\nu(\alpha R), \qquad \alpha_h^\dagger < \alpha. \tag{21}$$

The naive model overestimates range. The result does not require an isotropic smoother: anisotropy enters only through $a_k(e)$ and T_k, whose combination in (20) remains positive. For an isotropic kernel with $\Sigma_k = \sigma_k^2 I_d$,

$$A_{\nu,k}(r) = \sigma_k^2 \alpha^2 (2\nu + d - 2) G_\nu(z). \tag{22}$$

The qualitative fact that convolution can induce covariance anisotropy is known [4, 12]. The next corollary identifies the support-induced directional difference in the pairwise pseudo-inverse-range, including in the rough regimes where the leading common shift is not quadratic.

Corollary 1 (Directional support contrast). *For a unit vector e, let $\Delta_e(h) = \alpha - \alpha_h^\dagger(e)$, where the pair lag is $r = Re$ with the same fixed $R > 0$ in every direction. For two unit vectors e_1, e_2, under the assumptions of the phase theorem,*

$$\Delta_{e_1}(h) - \Delta_{e_2}(h) = \frac{\alpha^2 \{a_k(e_1) - a_k(e_2)\}}{R} \frac{K_{\nu-2}(z)}{K_{\nu-1}(z)} h^2 + o(h^2). \tag{23}$$

This formula holds for every fixed $\nu > 0$. For $0 < \nu < 1$ it is lower order than the common order-$h^{2\nu}$ range shift, and at $\nu = 1$ it is lower order than the common $h^2 \log(1/h)$ shift.

For a concrete two-dimensional family, let the base product Epanechnikov kernel be transformed according to

$$Z_{h,A}(t) = \int_{\mathbb{R}^2} k(u) Y(t - hAu)\, du, \qquad A_\varrho = \sqrt{\frac{2}{\varrho + \varrho^{-1}}}\, \mathrm{diag}(\sqrt{\varrho}, \varrho^{-1/2}), \tag{24}$$

where $\varrho \geq 1$. Then $\Sigma_k = A_\varrho A_\varrho^\top / 5$, its trace is fixed as ϱ varies, and its principal-variance ratio is ϱ^2. For $\nu > 1$, the ratio of the major- to minor-axis leading shift coefficients in (18) is

$$\frac{(2\nu - 1)\varrho^2 + 1}{\varrho^2 + 2\nu - 1}. \tag{25}$$

Thus elongated observation support can yield direction-dependent inferred range even though the latent covariance is isotropic. Equation (23), rather than this qualitative phenomenon, is the theorem-level statement used by the directional benchmark.

Remark 1 (Why symmetry does not imply an h^2 bias). Symmetry removes the order-h term at a fixed nonzero lag. When $0 < \nu < 1$, however, the origin expansion contains $\|r\|^{2\nu}$, which changes the variance at order $h^{2\nu}$. Normalizing by that variance dominates the order-h^2 off-diagonal change. At $\nu = 1$, the origin term is $h^2 \log(1/h)$. Only for $\nu > 1$ do ordinary second derivatives control both pieces.

Remark 2 (The fitted lag is fixed). The expansion holds for R bounded away from zero. If $R = ch$, the nonzero-lag Taylor expansion is no longer uniform and the relative target displacement need not vanish. No shrinking-lag conclusion is claimed.

4 Support-aware finite-grid likelihood

The continuous theorem concerns translation-invariant interior averaging. The finite experiment starts from raw sites $X_{\text{in}} = (x_1, \ldots, x_n)$ and a rectangular row-stochastic operator $S_h \in \mathbb{R}^{m \times n}$. If independent measurement noise is present before smoothing,

$$y = S_h\{Y(X_{\rm in}) + \epsilon\}, \qquad \epsilon \sim N(0, \tau^2 I_n), \tag{26}$$

then the correct covariance is

$$\Sigma_{\rm corr}(\alpha) = S_h\{K_\alpha(X_{\rm in}, X_{\rm in}) + \tau^2 I_n\} S_h^\top. \tag{27}$$

The naive declared-center point-support model instead uses

$$\Sigma_{\rm naive}(\alpha) = K_\alpha(X_{\rm out}, X_{\rm out}) + \tau^2 I_m. \tag{28}$$

Besides ignoring support, (28) incorrectly treats pre-smoothing noise as independent post-smoothing noise. With variance, smoothness, and nugget fixed at their generating values, the finite-design population target minimizes

$$Q_n(\alpha) = \frac{1}{2}\left[\log\det\Sigma(\alpha) + \mathrm{tr}\{\Sigma(\alpha)^{-1}\Sigma_0\}\right], \tag{29}$$

where constants independent of α are omitted. The corrected family contains Σ_0 exactly, provided the known operator and fixed nuisance parameters are correct and α is identifiable on the finite design; the naive family generally does not. Our primary support-only experiment sets $\tau = 0$, preventing noise timing from confounding the support effect. A separate combined-misspecification run with $\tau^2 = 0.05$ is retained as a reproducibility stress artifact but is not used for the principal claims.

A rectangular S_h is intentional. If a square parameter-independent smoother is invertible, the exact transformed likelihood is merely a change of coordinates and need not lose information. Rectangular aggregation represents the scientifically relevant reduction from raw locations to fewer supported measurements.

Boundary row normalization is not covered by the interior theorem. For two location-dependent scaled kernels with means μ_s and μ_t, a direct fixed-lag expansion gives

$$C_{h,s,t}(r) = C(r) + h\nabla C(r)^\top(\mu_s - \mu_t) + O(h^2). \tag{30}$$

A truncated kernel can therefore introduce an order-h term and destroy stationarity. The interior theorem provides no universal boundary sign. We report one boundary-retaining and one fixed jittered design as illustrations, not as a boundary scaling study.

4.1 Finite-library high-dimensional certificate

The replicated-field benchmark jointly varies variance and decay on a deterministic finite grid. The next proposition is a direct consequence of standard Gaussian quadratic-form concentration. It is included to state precisely what the benchmark certifies; it is not presented as a new concentration inequality.

Proposition 2 (Gaussian finite-library certificate)**.** *Let $X_1, \ldots, X_N \overset{\rm iid}{\sim} N_p(0, \Sigma_0)$, where Σ_0 is positive definite, and let $\{\Sigma_\theta : \theta \in \mathcal{G}\}$ be a deterministic library of $M < \infty$ positive-definite covariance matrices. Define*

$$\begin{aligned}
\widehat{\Sigma}_N &= \frac{1}{N}\sum_{i=1}^N X_i X_i^\top, \\
\widehat{L}_N(\theta) &= \frac{1}{2p}\left\{\log\det\Sigma_\theta + \mathrm{tr}(\Sigma_\theta^{-1}\widehat{\Sigma}_N)\right\}, \\
L(\theta) &= \mathbb{E}\widehat{L}_N(\theta), \qquad A_\theta = \Sigma_0^{1/2}\Sigma_\theta^{-1}\Sigma_0^{1/2}.
\end{aligned} \tag{31}$$

For any $\delta \in (0,1)$, put $t = \log(2M/\delta)$. With probability at least $1-\delta$, simultaneously for all $\theta \in \mathcal{G}$,

$$\left|\widehat{L}_N(\theta) - L(\theta)\right| \le \frac{\|A_\theta\|_{\mathrm{F}}}{p}\sqrt{\frac{t}{N}} + \frac{\|A_\theta\|_{\mathrm{op}}}{p}\frac{t}{N} =: u_\theta. \tag{32}$$

If $\widehat{\theta}$ and $\theta^\star$ minimize $\widehat{L}_N$ and L, respectively, over $\mathcal{G}$, then on the same event

$$L(\widehat{\theta}) - L(\theta^\star) \le 2\max_{\theta\in\mathcal{G}} u_\theta. \tag{33}$$

Moreover, $L(\theta)$ differs by a constant independent of θ from

$$\frac{1}{p} D_{\mathrm{KL}}\{N_p(0,\Sigma_0) \,\|\, N_p(0,\Sigma_\theta)\}\,. \tag{34}$$

No coordinate independence is assumed: all spatial dependence within one field is contained in Σ_0. Independence is required only across the N replicated fields. With quadratic effective rank

$$r_2(A_\theta) = \frac{\|A_\theta\|_{\mathrm{F}}^2}{\|A_\theta\|_{\mathrm{op}}^2},$$

the candidate radius is

$$\frac{\|A_\theta\|_{\mathrm{op}}}{p}\left\{\sqrt{\frac{r_2(A_\theta)t}{N}} + \frac{t}{N}\right\}.$$

Thus p helps only through controlled relative spectral geometry, not by itself. If constants $0 < c \le C < \infty$ satisfy

$$c\Sigma_0 \preceq \Sigma_\theta \preceq C\Sigma_0 \qquad \text{for every } \theta \in \mathcal{G}, \tag{35}$$

then $\|A_\theta\|_{\mathrm{op}} \le c^{-1}$ and $\|A_\theta\|_{\mathrm{F}} \le \sqrt{p}/c$, so

$$\max_\theta u_\theta \le \frac{1}{c}\left\{\sqrt{\frac{\log(2M/\delta)}{Np}} + \frac{\log(2M/\delta)}{Np}\right\}. \tag{36}$$

This is a conditional information rate, not a claim of separate fixed-domain consistency for Matérn variance and range. Relative spectral constants and criterion curvature can deteriorate with the design. The benchmark therefore records the actual Frobenius and operator norms and uses the exact radius in (32). A proof is given in Appendix C.

5 Synthetic validation

5.1 Continuous two-dimensional quadrature

We used a product Epanechnikov density

$$k(u) = \prod_{j=1}^{d} \frac{3}{4}(1-u_j^2)\mathbf{1}\{|u_j| \le 1\}. \tag{37}$$

The one-dimensional density of $U_j - V_j$ is available in closed form, so tensor Gauss–Legendre quadrature evaluates (4) deterministically. We used $d = 2$, $\alpha = R = 1$, quadrature order 96 per coordinate, 18 geometrically spaced bandwidths from 0.003 to 0.3, and $\nu \in \{0.25, 0.5, 0.75, 1, 1.5, 2.5\}$. For each cell we solved (8) by monotone root finding. Repeating the entire grid at quadrature orders

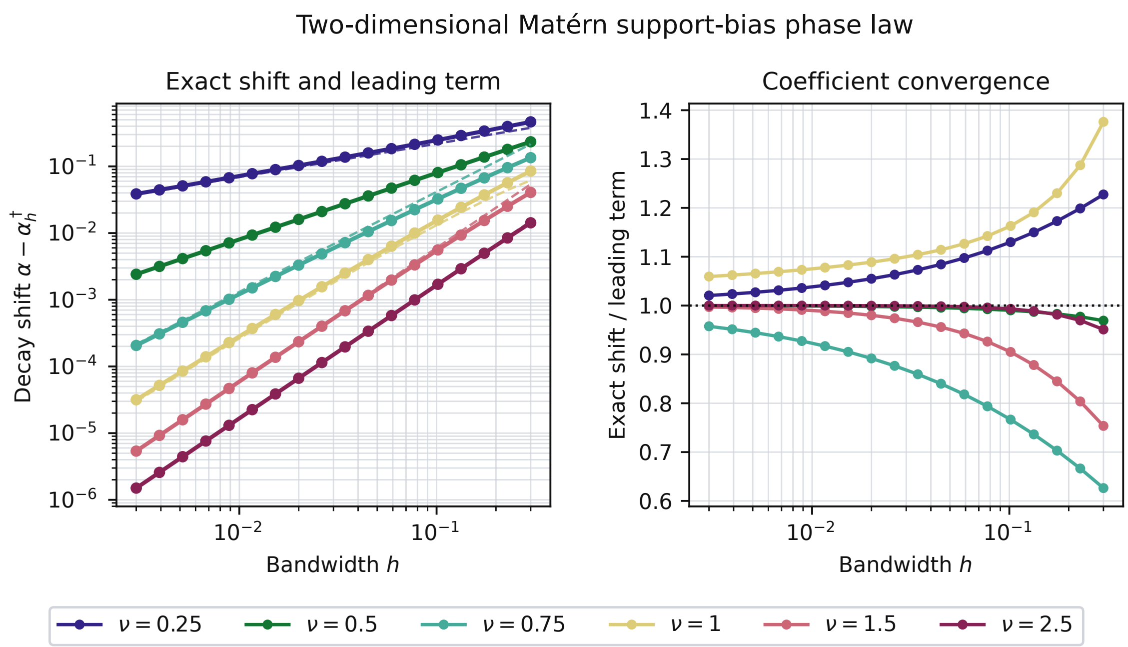

Figure 1: Continuous two-dimensional pseudo-decay shift under product Epanechnikov smoothing. Left: exact shifts (solid) and theorem leading terms with their explicit coefficients (dashed). Right: their ratio, which approaches one as $h \downarrow 0$.

64 and 128 changed pseudo-decay by at most 4.4×10^{-8} and 5.2×10^{-9}, respectively, relative to order 96. For this kernel $L = \sqrt{2}$, so the two largest displayed bandwidths, 0.229 and 0.3, are stress points outside the sufficient Taylor neighborhood $h \leq R/(4L)$. All slope and coefficient checks below use the six smallest bandwidths, which are inside that neighborhood.

All 108 computed decay shifts $\alpha - \alpha_h^\dagger$ were positive. An ordinary least-squares slope over the six smallest bandwidths on log–log axes was 0.515, 0.999, and 1.468 for $\nu = 0.25, 0.5, 0.75$, compared with theoretical powers 0.5, 1, and 1.5. At $\nu = 1$, the raw slope was 1.817, as expected from the logarithmic modifier rather than a pure power. The slopes for $\nu = 1.5$ and 2.5 were 1.994 and 2.000. Figure 1 displays the full paths and the coefficient-level theorem comparison. At the smallest bandwidth, exact-shift to leading-term ratios were 1.021, 1.000, 0.958, 1.059, 0.997, and 1.000 in increasing order of ν.

5.2 Directional support oracle

The directional track used (24) with $\varrho \in \{1, 4\}$, 19 lag angles from zero to 90 degrees, $\nu \in \{0.5, 1, 1.5, 2.5\}$, and 14 bandwidths from 0.003 to 0.15. All 2,128 computed inverse-range shifts were positive. At $\varrho = 4$, the major/minor smooth-regime coefficient ratios were 1.833 for $\nu = 1.5$ and 3.250 for $\nu = 2.5$, agreeing with (25). At $h = 0.15$, apparent range inflation along the major and minor support axes was 13.46% and 11.47% for $\nu = 0.5$, and 1.56% and 0.76% for $\nu = 1.5$.

Let D_ν denote the coefficient multiplying h^2 on the right side of (23) for the major-minus-minor comparison. At the smallest bandwidth, the absolute error in the ratio of the exact directional contrast to $D_\nu h^2$ was at most 4.3×10^{-6} over all four smoothnesses. Relative to quadrature order 96, orders 64 and 128 changed an individual shift by at most 1.3×10^{-8} of that shift and changed a directional contrast by at most 1.2×10^{-9} of that contrast.

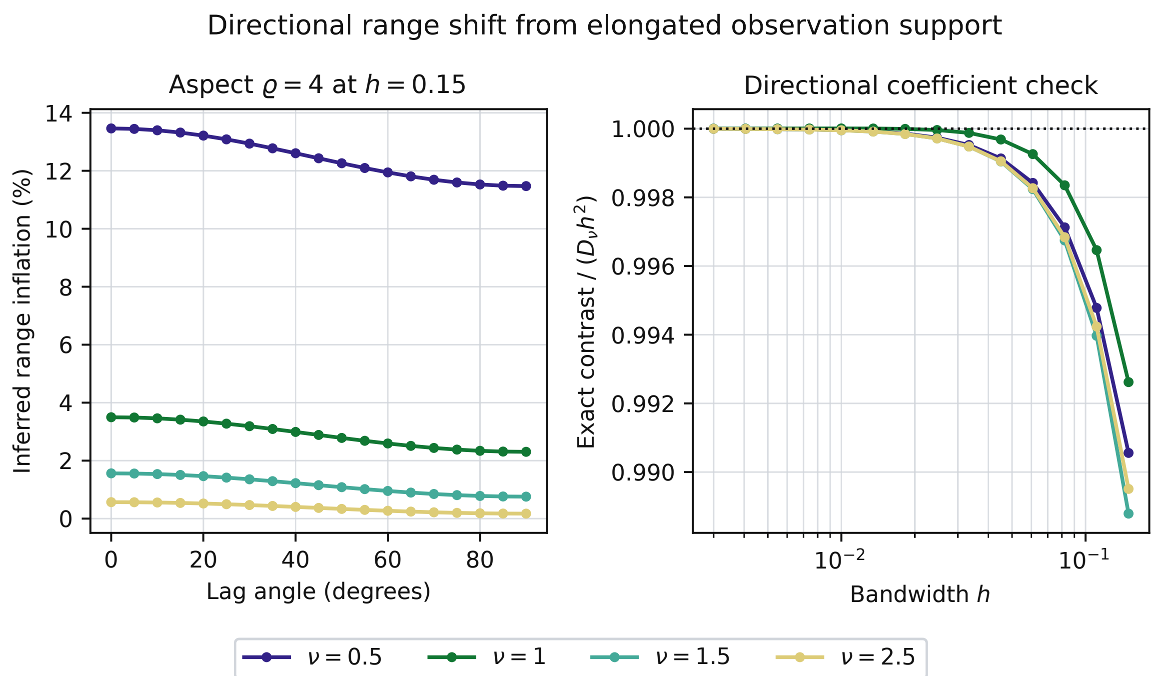

Figure 2: An isotropic latent field observed through elongated support. Left: inferred range inflation versus lag angle for aspect parameter $\varrho = 4$ and $h = 0.15$. Right: the exact major–minor inverse-range contrast divided by its theorem term $D_\nu h^2$.

5.3 Finite-grid full likelihood

The core design uses a 19×19 input lattice with spacing 0.25 and every second location as a candidate output center. Interior outputs are trimmed by 0.75 in each coordinate. We set $v = \alpha = 1$, $\tau^2 = 0$, $\nu \in \{0.5, 1, 1.5, 2.5\}$, and $h \in \{0, 0.3, 0.5, 0.7\}$. Radial Epanechnikov rows are normalized exactly. The zero-bandwidth operator is an exact rectangular selector. Stress panels retain boundary centers, perturb each input coordinate uniformly in $[-0.04, 0.04]$ before clipping to the original bounding box, and examine one-dimensional increasing-domain lattices with 100, 225, and 400 raw sites. The irregular panel uses one fixed jitter realization.

For each configuration, the finite-design population criterion is minimized by bounded scalar optimization over $\log \alpha \in [\log(0.02), \log(5)]$. Replicate likelihoods are evaluated on a 161-point equally spaced log-decay profile; an interior grid minimum is refined by three-point quadratic interpolation, while an endpoint minimum is retained as a boundary fit. Gaussian raw-observation vectors with the declared covariance are transformed by S_h. Configurations sharing all parameters except h reuse standard-normal draws. No numerical jitter is added. A Cholesky or population-optimization failure aborts its configuration and makes the reducer mark the run incomplete; no such failure occurred.

The promoted study used 200 replicates in each of 21 configurations, producing 8,400 fitted estimates. The reducer verified configuration hashes, row counts, and unique model–replicate keys for every task shard. Independent checks also verified finite numerical fields, expected model labels, replicate indices, and metadata. Across the 16 core two-dimensional configurations, the corrected population target differed from the true decay by at most 1.3×10^{-7}. At the largest bandwidth, $h = 0.7$, the naive population targets were 0.152, 0.414, 0.573, and 0.740 for $\nu = 0.5, 1, 1.5, 2.5$, respectively. Their corresponding Monte Carlo means were 0.151, 0.407, 0.570, and 0.738; the

Table 1: Finite-design validation at $h = 0.7$. The table is generated from immutable task shards; no failed or boundary-hitting fit is silently removed.

ν	Model	Target	Mean	MCSE	Bound hits
0.5	Corrected	1.000	1.090	0.047	3
0.5	Naive	0.152	0.151	0.004	0
1	Corrected	1.000	0.990	0.010	0
1	Naive	0.414	0.407	0.005	0
1.5	Corrected	1.000	0.990	0.007	0
1.5	Naive	0.573	0.570	0.005	0
2.5	Corrected	1.000	0.994	0.004	0
2.5	Naive	0.740	0.738	0.003	0

corrected means were 1.090, 0.990, 0.990, and 0.994. The comparatively noisy corrected estimate at $\nu = 0.5$ nevertheless remained within two Monte Carlo standard errors of its population target.

No optimizer or covariance failure was discarded. Six of the 6,400 core fits landed at the declared bounds, and those fits remain in Figure 3 and Table 1. Including boundary outputs changed both the output dimension and the rough-field naive target, from 0.152 to 0.135 at $h = 0.7$; for the fixed jittered design the target changed from 0.289 to 0.284 at $h = 0.5$. In both stress cases the corrected target remained one. In the one-dimensional domain-growth check, increasing the number of raw sites from 100 to 400 reduced the corrected estimator standard deviation from 0.529 to 0.177 and the naive standard deviation from 0.151 to 0.068, while at $n = 400$ their means remained within 0.026 and 0.009, respectively, of their population targets (Figure 4).

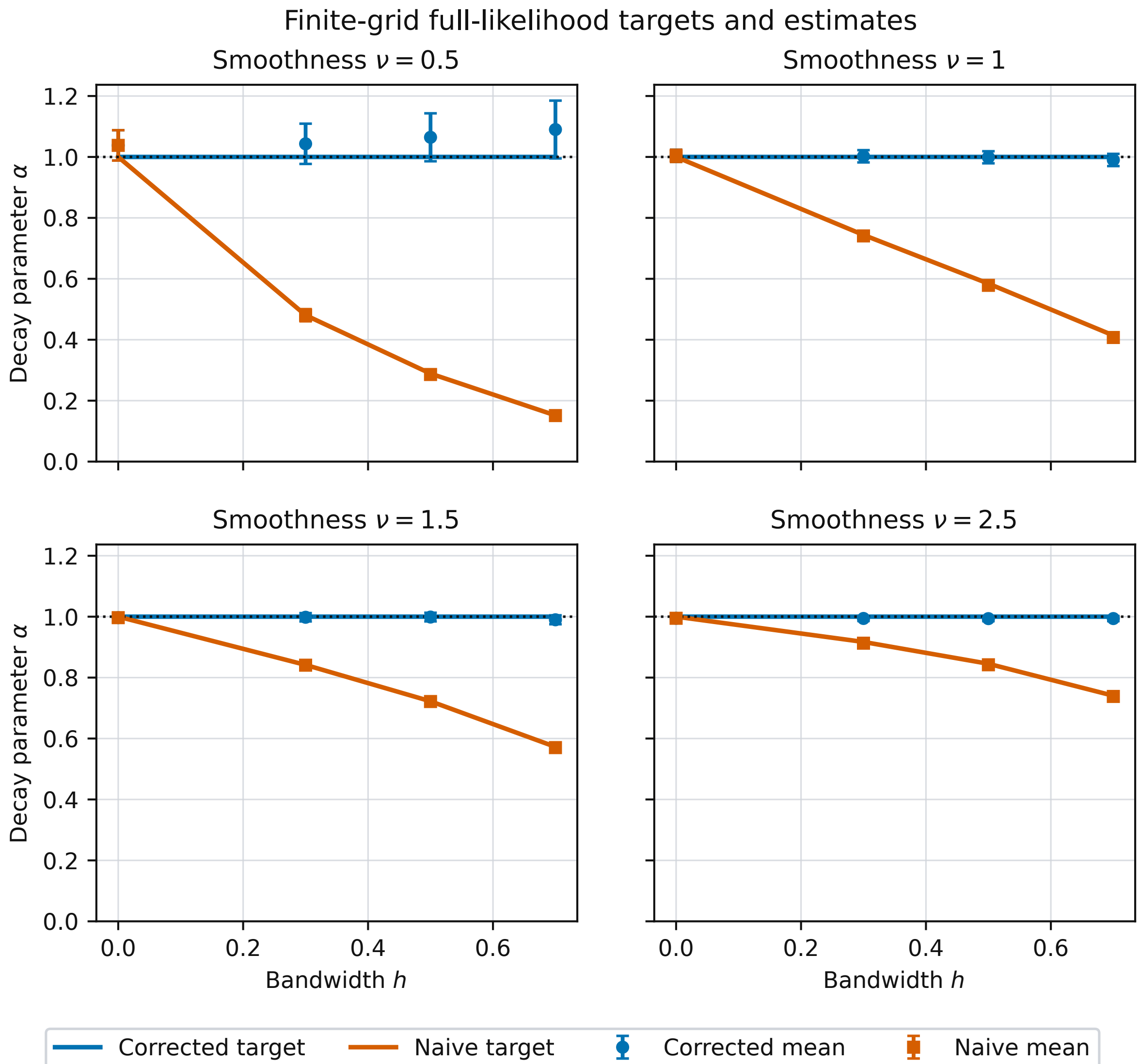

Figure 3: Finite-design population targets and Monte Carlo mean decay estimates for corrected and naive likelihoods. Error bars show two Monte Carlo standard errors of the mean, not estimator confidence intervals.

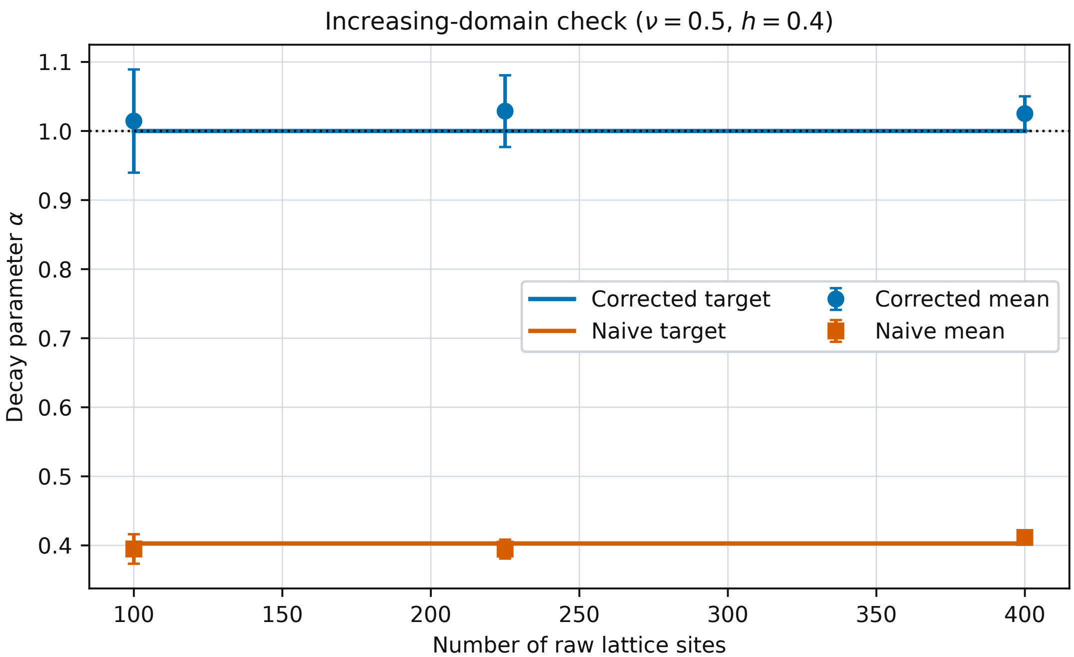

Figure 4: Increasing-domain check for the finite-grid likelihood. Lines are nonrandom finite-design targets and points are Monte Carlo means from 200 replicates; error bars show two Monte Carlo standard errors of each mean.

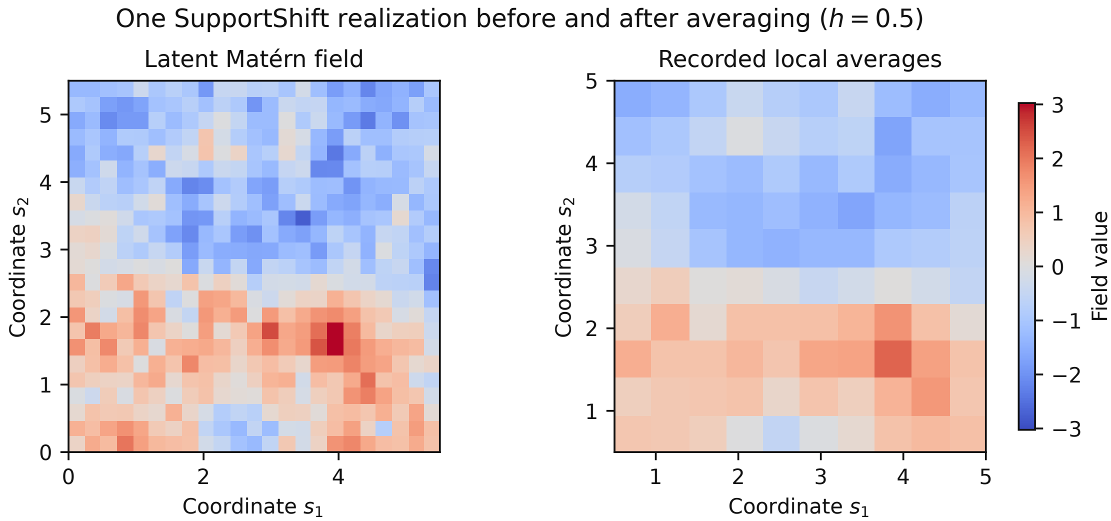

Figure 5: One synthetic field before and after the known rectangular observation operator: a 23×23 latent realization (left) and its 10×10 locally averaged observations (right). This illustration is generated from the same covariance and operator implementation as the likelihood benchmark.

5.4 Replicated-field finite-library experiment

To connect the probability statement to an estimator, we generated independent replicated spatial fields on increasing-domain two-dimensional designs at fixed lattice spacing. For $q \in \{4, 6, 8, 10\}$, a padded $(2q + 3) \times (2q + 3)$ latent lattice with spacing 0.25 was mapped by a full-row-rank radial Epanechnikov operator of bandwidth 0.5 to a $q \times q$ output lattice, giving $p = q^2 \in \{16, 36, 64, 100\}$. The mean was known to be zero. We used $\nu \in \{0.5, 1.5\}$, $v = \alpha = 1$, and $N \in \{1, 4, 16, 64\}$ independent fields. The candidate set was an explicit deterministic $101 \times 161 = 16{,}261$ Cartesian library of variance and inverse-range values spanning $[0.35, 2.5] \times [0.15, 1.6]$ and containing the generating parameter pair $(1, 1)$ exactly. This truth anchoring is an oracle benchmark control, not a deployable rule for choosing a data-analysis grid. The complete library was fixed before the final Monte Carlo run. Corrected candidates were $vS_hR_\alpha S_h^\top$; naive candidates were vR_α at the declared output centers. Thus only the corrected covariance library contained the data law. Every empirical minimizer was obtained by exact finite-library enumeration, without continuous interpolation.

For each (p, ν) cell, 200 nested replicate-field sequences were generated from the support-aware truth. Thus there were 64 distinct model–p–ν–N certificate cells and 12,800 fitted estimates. The confidence parameter was $\delta = 0.05$, and the theorem radius was computed from the actual candidate-specific Frobenius and operator norms. No independence approximation was made for coordinates within a field. The certificate was applied separately to each fixed design, model, smoothness, and sample size; a joint 95% statement over all 64 reported cells is not claimed. The raw illustration in Figure 5 shows the observation map used for the largest design.

The discrete population objective was within 2.63×10^{-5} per coordinate of the continuously profiled objective in all 16 design–model cases, below the 5×10^{-5} tolerance fixed before the final Monte Carlo run. In every one of the 64 model–p–ν–N cells, all 200 trials satisfied the actual simultaneous candidatewise event

$$|\widehat{L}_N(\theta) - L(\theta)| \le u_\theta \quad \text{for every } \theta \in \mathcal{G}.$$

Over all 12,800 fits, the largest value of $\max_\theta |\widehat{L}_N(\theta) - L(\theta)|/u_\theta$ was 0.789. The weaker worst-radius

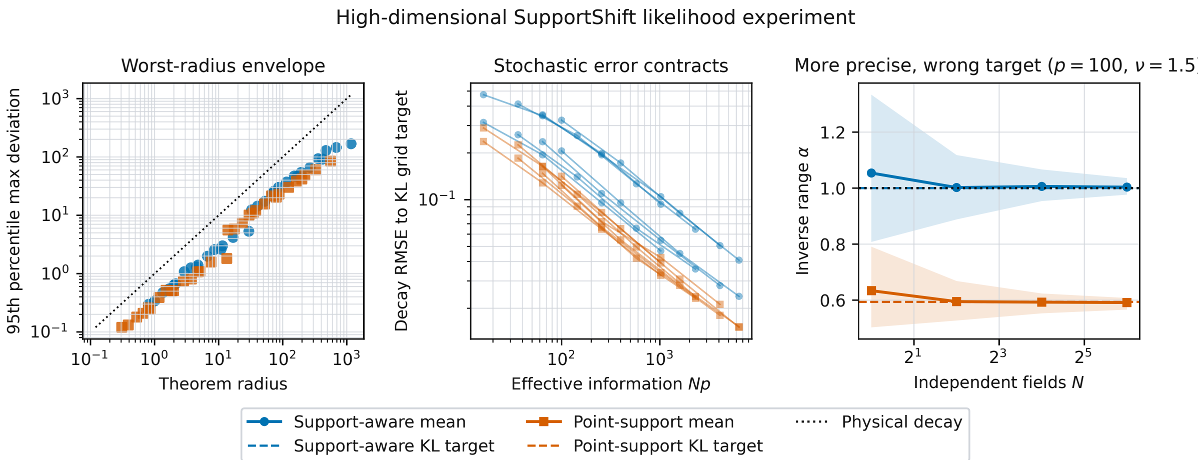

Figure 6: The replicated-field finite-library benchmark. Left: empirical 95th percentiles of the worst likelihood deviation versus the worst-candidate theorem radius; exact candidatewise coverage is reported in the text. Center: inverse-range RMSE to each model's finite-grid KL target versus Np. Right: for $p = 100$ and $\nu = 1.5$, increasing N concentrates the corrected estimator near the physical truth and the naive estimator near its displaced target.

envelope displayed in Figure 6 also held throughout: across cells, the 95th percentile of $\max_\theta |\widehat{L}_N - L|$ was at most 0.402 times $\max_\theta u_\theta$, with median ratio 0.298. These 100% empirical coverages show that the finite-library union bound was conservative here; each 95% certificate applies to its own fixed cell, not jointly across all 64.

For each fixed model, dimension, and smoothness, a log–log regression of the 95th percentile criterion deviation on N produced slopes from -0.528 to -0.454, with median -0.505, matching the $N^{-1/2}$ leading dependence in (32). The exact deterministic ERM inequality $L(\widehat{\theta}) - L(\theta^\star) \leq 2\max_\theta |\widehat{L}_N(\theta) - L(\theta)|$ held for every fitted trial.

The support-aware library had population target $(v, \alpha) = (1, 1)$ in every design. In contrast, at $p = 100$ the naive finite-grid inverse-range targets were 0.320 for $\nu = 0.5$ and 0.594 for $\nu = 1.5$, close to continuously profiled targets 0.318 and 0.589. For $p = 100, \nu = 1.5$, increasing N from 1 to 64 reduced support-aware decay RMSE from 0.206 to 0.024 and naive RMSE to its own KL target from 0.124 to 0.015. Yet the naive RMSE to the physical inverse range was 0.409 at $N = 64$, and its mean estimate was 0.591. The rough case was more pronounced: at $p = 100, \nu = 0.5, N = 64$, naive RMSE was 0.015 to its target but 0.681 to the physical value. Thus sampling spread decreased while support misspecification remained. Figure 6 visualizes this "more information, more confidently wrong" distinction. The parameter-error panels are empirical summaries, not consequences of Proposition 2, which controls criterion deviation and excess risk unless additional curvature is imposed.

6 Discussion

The theorem separates two effects that are often conflated. Linear propagation of covariance through a known support is standard. The new object here is the parameter selected when that propagation is ignored. For rough Matérn fields, the origin singularity makes the pseudo-target move at a fractional power of bandwidth. For smooth fields, a Bessel recurrence shows that the combined variance and nonzero-lag correction still increases normalized correlation, including under anisotropic smoothing.

The simulations have a deliberate role: they validate theorem-linked generators, compute nonrandom finite-design KL targets, and distinguish misspecification from finite-sample error. They are not substitutes for the phase-law proof. The zero-support case verifies exact agreement, the corrected target verifies model containment, and the boundary and irregular panels expose where the interior theorem stops. The directional oracle checks a coefficient rather than only a sign. The replicated-field experiment separately checks the finite-library probability event and demonstrates why shrinking sampling error does not repair a displaced population target.

Several limitations define the scope of the result. First, ν, the support, and the noise timing are treated as known. Jointly estimating smoothness, bandwidth, variance, and range may be weakly identified. Second, the closed target is for a two-point composite fit. A full likelihood cannot generally match every smoothed lag; its KL target is computed numerically in our finite designs. Third, fixed-resolution discrete averaging does not converge to continuous convolution merely because the spatial domain grows. A mixed-domain quadrature limit or an exact discrete-weight theorem is required. Fourth, boundary normalization creates location-dependent supports and can introduce first-order terms. Fifth, the original finite-design validation fixes nuisance parameters to isolate the target shift. The replicated-field track jointly varies variance and decay, but only over a declared finite library; it does not establish a continuous-parameter rate. Sixth, the exact certificate radius uses the generating Σ_0. It is an oracle simulation certificate, not a plug-in confidence procedure for unknown real-data covariance.

These restrictions are preferable to an unsupported broad claim. They also suggest direct extensions: a uniform discrete-to-continuous approximation, continuous joint nuisance profiling, spectral full-likelihood expansions, curvature conditions translating likelihood error into parameter error, and explicit boundary counterexamples. The finite-library result gives a useful high-dimensional probability framing, but the concentration step itself is standard supporting machinery. The paper's novelty rests on the explicit support-distortion phase law and directional coefficient, not on a new quadratic-form inequality.

7 Conclusion

For a fixed nonzero-lag pair fit under translation-invariant interior averaging, ignored support makes a Matérn field look more persistent than it is. The magnitude is controlled by smoothness: $h^{2\nu}$, $h^2 \log(1/h)$, or h^2, while anisotropic support produces an explicit order-h^2 contrast between lag directions. A support-aware observation covariance removes the finite-design population shift when the operator is known, nuisance parameters are correctly specified, and decay is identifiable. For a deterministic covariance library fitted to independent replicated fields, a simultaneous Gaussian likelihood certificate quantifies the sampling-error component without assuming independent spatial coordinates. Together, these results diagnose both the population displacement and the finite-sample concentration around it; they make no universal claim for boundary supports or arbitrary continuous full-likelihood fits.

A Proof of the phase theorem

Proof. Let $f_\alpha(x) = \mathcal{M}_\nu(\alpha\|x\|)$. Since r is bounded away from zero and D is bounded, a fourth-order Taylor expansion is uniform. Because $D = U - V$ has a centrally symmetric law, both $\mathbb{E}D = 0$ and $\mathbb{E}(D^{\otimes 3}) = 0$; moreover $\mathbb{E}(DD^\top) = 2\Sigma_k$. Therefore

$$\mathbb{E}f_\alpha(r + hD) = f_\alpha(r) + \frac{h^2}{2}\operatorname{tr}\{\nabla^2 f_\alpha(r)\mathbb{E}(DD^\top)\} + O(h^4). \tag{38}$$

For $r = Re$,

$$\nabla^2 f_\alpha(r) = \alpha^2\mathcal{M}_\nu''(z)ee^\top + \alpha^2\frac{\mathcal{M}_\nu'(z)}{z}(I - ee^\top), \tag{39}$$

which proves (11).

For noninteger ν, the origin expansion is

$$\mathcal{M}_\nu(x) = \sum_{j=0}^{\infty}\frac{(x^2/4)^j}{j!(1-\nu)_j} + \frac{\Gamma(-\nu)}{2^{2\nu}\Gamma(\nu)}x^{2\nu}\sum_{j=0}^{\infty}\frac{(x^2/4)^j}{j!(1+\nu)_j}. \tag{40}$$

Taking expectations at $x = \alpha h\|D\|$ yields (13) and (15). At $\nu = 1$,

$$\mathcal{M}_1(x) = 1 + \frac{x^2}{2}\{\log(x/2) + \gamma_{\mathrm{E}} - 1/2\} + O\{x^4(1 + |\log x|)\}, \tag{41}$$

which proves (14).

At the second transition,

$$\mathcal{M}_2(x) = 1 - \frac{x^2}{4} + x^4\left\{\frac{3}{64} - \frac{\log(x/2) + \gamma_{\mathrm{E}}}{16}\right\} + O\{x^6(1 + |\log x|)\}. \tag{42}$$

This gives an $O\{h^4\log(1/h)\}$ variance remainder at $\nu = 2$. For fixed $\nu > 2$, four-times differentiability at the origin gives

$$\mathcal{M}_\nu(x) = 1 - \frac{x^2}{4(\nu-1)} + O(x^4).$$

For $1 < \nu < 2$, the fractional term in (40) is $O(x^{2\nu})$.

Divide the fixed-lag expansion by the relevant variance expansion and apply a first-order Taylor expansion to the inverse of $\alpha \mapsto \mathcal{M}_\nu(\alpha R)$. This directly gives (16) and (17); the same division and inverse expansion preserve the stated $O(h^{2\nu})$, $O\{h^4\log(1/h)\}$, and $O(h^4)$ remainders in the three smooth subregimes. For $\nu > 1$, the order-h^2 coefficient of normalized correlation is

$$B_{\nu,k}(r) + \mathcal{M}_\nu(z)\frac{\alpha^2 m_2}{4(\nu-1)}. \tag{43}$$

The Matérn correlation satisfies

$$\mathcal{M}_\nu''(z) + \frac{1-2\nu}{z}\mathcal{M}_\nu'(z) - \mathcal{M}_\nu(z) = 0. \tag{44}$$

Using $m_2 = 2T_k$, the coefficient becomes $A_{\nu,k}(r)$. Finally,

$$K_\nu(z) = K_{\nu-2}(z) + \frac{2(\nu-1)}{z}K_{\nu-1}(z) \tag{45}$$

implies (20). Positivity of $K_\eta(z)$ for real η and $z > 0$, together with (9), proves the sign. □

B Proof of the directional support contrast

Proof. For brevity, write $B_e = B_{\nu,k}(Re)$. The fixed-lag expansion gives

$$\frac{C_h(Re)}{v} = \mathcal{M}_\nu(z) + h^2 B_e + O(h^4). \tag{46}$$

The zero-lag denominator $d_h = C_h(0)/v$ is common to all lag directions and satisfies $d_h \to 1$. Consequently,

$$\rho_h(Re_1) - \rho_h(Re_2) = h^2\{B_{e_1} - B_{e_2}\} + o(h^2). \tag{47}$$

This cancellation is why the statement remains order h^2 when the common rough-field shift contains lower powers of h.

Let $g(s) = \mathcal{M}_\nu^{-1}(s)/R$ on a neighborhood of $\mathcal{M}_\nu(z)$. Since $\mathcal{M}_\nu'(z) < 0$, the inverse-function theorem gives $g'\{\mathcal{M}_\nu(z)\} = 1/\{R\mathcal{M}_\nu'(z)\}$. Also $\rho_h(Re_j) \to \mathcal{M}_\nu(z)$, so the mean-value theorem and (47) imply

$$\Delta_{e_1}(h) - \Delta_{e_2}(h) = -\frac{B_{e_1} - B_{e_2}}{R\mathcal{M}_\nu'(z)} h^2 + o(h^2). \tag{48}$$

From (10),

$$B_{e_1} - B_{e_2} = \alpha^2\{a_k(e_1) - a_k(e_2)\}\left\{\mathcal{M}_\nu''(z) - \frac{\mathcal{M}_\nu'(z)}{z}\right\}. \tag{49}$$

If $c = 2^{1-\nu}/\Gamma(\nu)$, the Bessel derivative identity yields

$$\mathcal{M}_\nu'(z) = -cz^\nu K_{\nu-1}(z), \qquad \mathcal{M}_\nu''(z) - \frac{\mathcal{M}_\nu'(z)}{z} = cz^\nu K_{\nu-2}(z). \tag{50}$$

Substitution into (48) proves (23). The identities and positivity of $K_\eta(z)$ hold for every real order η and $z > 0$, so no condition $\nu > 1$ is needed for the directional contrast. $\square$

C Proof of the finite-library certificate

Proof. Write $X_i = \Sigma_0^{1/2} Z_i$, where the Z_i are independent $N_p(0, I_p)$ vectors, and stack them into $Z = (Z_1^\top, \ldots, Z_N^\top)^\top$. For a fixed candidate θ,

$$\sum_{i=1}^N X_i^\top \Sigma_\theta^{-1} X_i = Z^\top (I_N \otimes A_\theta) Z, \qquad \mathbb{E} Z^\top (I_N \otimes A_\theta) Z = N \operatorname{tr}(A_\theta). \tag{51}$$

The two-sided Gaussian quadratic-form inequality gives, for every $t > 0$,

$$\Pr\left[\left|Z^\top (I_N \otimes A_\theta) Z - N \operatorname{tr}(A_\theta)\right| > 2\sqrt{N}\|A_\theta\|_{\mathrm{F}}\sqrt{t} + 2\|A_\theta\|_{\mathrm{op}} t\right] \le 2e^{-t}. \tag{52}$$

Here $\|I_N \otimes A_\theta\|_{\mathrm{F}} = \sqrt{N}\|A_\theta\|_{\mathrm{F}}$ and $\|I_N \otimes A_\theta\|_{\mathrm{op}} = \|A_\theta\|_{\mathrm{op}}$. The log-determinant term in $\widehat{L}_N(\theta) - L(\theta)$ is deterministic. Dividing (52) by $2Np$, setting $t = \log(2M/\delta)$, and applying a union bound over the deterministic library proves (32) [9, 11].

On the simultaneous event,

$$\begin{aligned} L(\widehat{\theta}) - L(\theta^\star) &= [L - \widehat{L}_N](\widehat{\theta}) + [\widehat{L}_N(\widehat{\theta}) - \widehat{L}_N(\theta^\star)] + [\widehat{L}_N - L](\theta^\star) \\ &\le u_{\widehat{\theta}} + u_{\theta^\star} \le 2 \max_{\theta \in \mathcal{G}} u_\theta, \end{aligned} \tag{53}$$

which proves (33). Finally,

$$D_{\mathrm{KL}}\{N_p(0,\Sigma_0) \,\|\, N_p(0,\Sigma_\theta)\} = \frac{1}{2}\left[\mathrm{tr}(\Sigma_\theta^{-1}\Sigma_0) - p + \log\frac{\det\Sigma_\theta}{\det\Sigma_0}\right], \tag{54}$$

which proves (34). Under (35), inversion and congruence give $0 \prec A_\theta \preceq c^{-1}I_p$; hence the stated operator- and Frobenius-norm bounds and (36) follow. □

D Exact exponential benchmark

For $d = 1$, $\nu = 1/2$, and the Epanechnikov density $k(u) = \frac{3}{4}(1-u^2)\mathbf{1}\{|u| \le 1\}$, the density of $D = U - V$ is

$$g(w) = \frac{3}{5} - \frac{3}{4}|w|^2 + \frac{3}{8}|w|^3 - \frac{3}{160}|w|^5, \qquad |w| \le 2. \tag{55}$$

With $x = \alpha h$, define

$$s(x) = \mathbb{E}e^{-x|D|}, \qquad M_k(x) = \mathbb{E}e^{xU} = \frac{3(x\cosh x - \sinh x)}{x^3}, \qquad q(x) = M_k(x)^2. \tag{56}$$

Then $C_h(0) = vs(x)$, and for $R \ge 2h$,

$$C_h(R) = vq(x)e^{-\alpha R}. \tag{57}$$

The exact naive pair target is

$$v_h^\dagger = vs(x), \qquad \alpha_h^\dagger = \alpha - \frac{1}{R}\log\frac{q(x)}{s(x)}. \tag{58}$$

For $x > 0$, compact support and nondegeneracy give

$$1 < q(x)/s(x) < e^{2x}.$$

Indeed, condition on $A = |U|$ and $B = |V|$, and put $p = A+B$ and $m = |A-B|$. Symmetry of the signs gives conditional numerator and denominator proportional to $\cosh(xp)+\cosh(xm)$ and $e^{-xp}+e^{-xm}$, respectively. The lower inequality follows from $\cosh(t) \ge e^{-t}$, with strictness off $A = B = 0$. Since $p + m = 2\max(A,B) \le 2$, $\cosh(xp) < e^{xp} \le e^{2x-xm}$ and $\cosh(xm) \le e^{xm} \le e^{2x-xp}$, with strictness on a set of positive probability. Averaging proves the upper inequality. Since $R \ge 2h$, this implies $0 < \alpha - R^{-1}\log\{q(x)/s(x)\} < \alpha$. Variance and decay are therefore both strictly underestimated. The small-bandwidth expansions are

$$s(x) = 1 - \frac{18}{35}x + \frac{1}{5}x^2 - \frac{4}{63}x^3 + \frac{3}{175}x^4 + O(x^5), \tag{59}$$

$$q(x) = 1 + \frac{1}{5}x^2 + \frac{3}{175}x^4 + O(x^6), \tag{60}$$

$$\log\{q(x)/s(x)\} = \frac{18}{35}x + \frac{162}{1225}x^2 + O(x^3). \tag{61}$$

This benchmark is used as an exact software regression test, not as the paper's novelty claim.

E Reproducibility contract

The public code at https://github.com/Sasha-Cui/HighDimSpatialStatistics records the exact configuration and commit identifier in each promoted run's metadata; the frozen paper artifact is tagged `supportshift-geosim-v1.0.0`. The finite-grid full-likelihood track uses an immutable JSON manifest, SHA-256 configuration hashes, deterministic seed derivation, one task-local atomic shard, and an `afterany` reducer that refuses to mark a run complete when a shard is missing or invalid. The replicated-field track records its complete candidate grids, matrix-norm radii, derived seeds, dependency versions, result hashes, and validation gates. Those gates check the exact finite-grid ERM inequality, agreement with a public likelihood implementation, population-target interiority, finite-to-continuous objective resolution, the candidatewise simultaneous event, and exact reapplication of the saved observation operator. The promoted batch aborts unless the Git worktree is clean. The zero-bandwidth selector, row sums, row rank, corrected population target, grid interpolation, Gaussian certificate, driver shakedown, artifact hash manifest, and closed exponential formulas are tested. The reported implementation adds no adaptive diagonal jitter. All paper figures and tables are regenerated from audited aggregate CSV files.

References

[1] Takeshi Amemiya and Roland Y. Wu. The effect of aggregation on prediction in the autoregressive model. *Journal of the American Statistical Association*, 67(339):628–632, 1972. doi: 10.1080/01621459.1972.10481264.

[2] François Bachoc. Asymptotic analysis of covariance parameter estimation for Gaussian processes in the misspecified case. *Bernoulli*, 24(2):1531–1575, 2018. doi: 10.3150/16-BEJ906.

[3] Erick A. Chacón-Montalván, Peter M. Atkinson, Christopher Nemeth, Benjamin M. Taylor, and Paula Moraga. Spatial latent Gaussian modelling with change of support, 2024.

[4] David Clifford, Alex B. McBratney, James Taylor, and Brett M. Whelan. Generalized analysis of spatial variation in yield monitor data. *The Journal of Agricultural Science*, 144(1):45–51, 2006. doi: 10.1017/S0021859606005892.

[5] Maria Myrto Folia and Magnus Rattray. Trajectory inference and parameter estimation in stochastic models with temporally aggregated data. *Statistics and Computing*, 28:1053–1072, 2018. doi: 10.1007/s11222-017-9779-x.

[6] Montserrat Fuentes. Approximate likelihood for large irregularly spaced spatial data. *Journal of the American Statistical Association*, 102(477):321–331, 2007. doi: 10.1198/016214506000000852.

[7] Alan E. Gelfand, Li Zhu, and Bradley P. Carlin. On the change of support problem for spatio-temporal data. *Biostatistics*, 2(1):31–45, 2001. doi: 10.1093/biostatistics/2.1.31.

[8] Carol A. Gotway and Linda J. Young. Combining incompatible spatial data. *Journal of the American Statistical Association*, 97(458):632–648, 2002. doi: 10.1198/016214502760047140.

[9] Daniel Hsu, Sham M. Kakade, and Tong Zhang. A tail inequality for quadratic forms of subgaussian random vectors. *Electronic Communications in Probability*, 17:1–6, 2012. doi: 10.1214/ECP.v17-2079.

[10] Phaedon C. Kyriakidis and Eun-Hye Yoo. Geostatistical prediction and simulation of point values from areal data. *Geographical Analysis*, 37(2):124–151, 2005. doi: 10.1111/j.1538-4632.2005.00633.x.

[11] Béatrice Laurent and Pascal Massart. Adaptive estimation of a quadratic functional by model selection. *The Annals of Statistics*, 28(5):1302–1338, 2000. doi: 10.1214/aos/1015957395.

[12] Christopher J. Paciorek and Mark J. Schervish. Spatial modelling using a new class of nonstationary covariance functions. *Environmetrics*, 17(5):483–506, 2006. doi: 10.1002/env.785.

[13] Frederik J. Simons, Olivia L. Walbert, Arthur P. Guillaumin, Gabriel L. Eggers, Kevin W. Lewis, and Sofia C. Olhede. Maximum-likelihood estimation of the Matérn covariance structure of isotropic spatial random fields on finite, sampled grids. *Geophysical Journal International*, 245(2):ggag044, 2026. doi: 10.1093/gji/ggag044.

[14] Michael L. Stein. *Interpolation of Spatial Data: Some Theory for Kriging*. Springer, New York, 1999. doi: 10.1007/978-1-4612-1494-6.

[15] Daniel O. Stram and William W. S. Wei. Temporal aggregation in the ARIMA process. *Journal of Time Series Analysis*, 7(4):279–292, 1986. doi: 10.1111/j.1467-9892.1986.tb00495.x.

[16] Hao Zhang. Inconsistent estimation and asymptotically equal interpolations in model-based geostatistics. *Journal of the American Statistical Association*, 99(465):250–261, 2004. doi: 10.1198/016214504000000241.